

Frequency-Domain Waveform Emulation for Gravitational Waves

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Abstract

We present a framework for neural network emulation of gravitational waveforms in the frequency domain, the natural setting for matched-filter data analysis. The frequency domain offers significant simplifications: extrinsic parameters enter analytically, the mass-scaled frequency Mf provides a universal domain variable, and non-precessing waveforms reduce to smooth amplitude and phase functions without coprecessing frame complications. We detail representation choices, symmetries, and architectural options for neural network emulation, with particular attention to the aligned-spin higher-mode case relevant for IMR-PhenomXHM.

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1 Introduction

Gravitational wave data analysis operates primarily in the frequency domain. The fundamental detection statistic—the matched filter—is computed as a frequency-domain inner product, and detector noise is characterised by its power spectral density $S_n(f)$. Frequency-domain waveform models (IMRPhenomX, IMRPhenomT, etc.) are the workhorses of LIGO-Virgo-KAGRA parameter estimation.

This document presents a framework for neural network emulation directly in the frequency domain. Compared to time-domain emulation, the frequency domain offers several advantages:

- **Mass scaling:** Waveforms depend on the dimensionless product Mf , not mass and frequency separately. This dramatically simplifies the parameter space.
- **Analytic extrinsic dependence:** Distance, coalescence time, and reference phase enter as simple multiplicative factors, eliminating them from the network’s learning task.
- **No coprecessing dynamics:** For non-precessing systems, there are no time-evolving Euler angles—just smooth amplitude and phase functions.
- **Direct integration with data analysis:** Emulated waveforms can be used directly in likelihood evaluations without Fourier transformation.

For precessing systems, the frequency domain is less natural—precession dynamics are inherently time-domain phenomena that are approximated in frequency-domain models via the “twisting-up” procedure. We discuss this and its implications in §7.

The primary target of this document is emulation of the IMRPhenomXHM family: non-precessing (aligned-spin) waveforms with higher harmonics, computed in the frequency domain.

2 Conventions

Complex strain. The gravitational wave strain is

$$h \equiv h_+ - ih_\times \quad (1)$$

and its Fourier transform is

$$\tilde{h}(f) = \int_{-\infty}^{\infty} h(t) e^{-2\pi i f t} dt \quad (2)$$

We work with positive frequencies $f > 0$ throughout; negative frequencies are related by conjugate symmetry of the real time-domain signal.

Geometric units. We use units with $G = c = 1$, so mass has dimensions of time (or length). The total mass $M = m_1 + m_2$ sets the characteristic timescale. Define the solar mass timescale:

$$\tau_\odot \equiv \frac{GM_\odot}{c^3} \approx 4.926 \times 10^{-6} \text{ s} \quad (3)$$

For a binary with total mass M (expressed in solar masses), the geometric timescale is $M\tau_\odot$. Physical frequency is recovered from dimensionless frequency Mf via:

$$f_{\text{phys}} = \frac{Mf}{M\tau_\odot} = \frac{Mf}{M/M_\odot} \times \frac{1}{\tau_\odot} \approx \frac{Mf}{M/M_\odot} \times 2.03 \times 10^5 \text{ Hz} \quad (4)$$

Dimensionless frequency. The key insight for emulation is that gravitational waveforms depend on the *dimensionless* combination

$$\boxed{\mathcal{F} \equiv Mf} \quad (5)$$

rather than M and f separately. This follows from the scale-invariance of general relativity: there is no intrinsic mass scale in vacuum GR. All time/frequency scales are set by the total mass M .

Geometric frequency range. In geometric units, typical LIGO sources span:

- Inspiral onset: $\mathcal{F} \sim 10^{-4}$ to 10^{-3}
- Merger: $\mathcal{F} \sim 0.01$ to 0.02
- Ringdown: $\mathcal{F} \sim 0.05$ to 0.1

A single emulator covering $\mathcal{F} \in [10^{-4}, 0.3]$ handles all masses from $1M_\odot$ to $1000M_\odot$ across the full LIGO band.

Spin-weighted spherical harmonics. We use the standard convention for ${}^{-2}Y_{\ell m}(\iota, \varphi_0)$ consistent with LALSuite and [1]. Under rotation, modes mix according to the Wigner D-matrices.

Reference frequency. Waveforms are typically specified at a reference frequency f_{ref} where the orbital phase ϕ_{ref} is defined. In terms of $\mathcal{F}$, this becomes $\mathcal{F}_{\text{ref}} = Mf_{\text{ref}}$.

3 Frequency-Domain Decomposition

3.1 Spherical Harmonic Expansion

The frequency-domain strain observed at inclination ι and reference azimuth φ_0 is:

$$\boxed{\tilde{h}(f; \boldsymbol{\lambda}; \iota, \varphi_0) = \sum_{\ell=2}^{\ell_{\text{max}}} \sum_{m=-\ell}^{+\ell} \tilde{h}_{\ell m}(f; \boldsymbol{\lambda}) {}^{-2}Y_{\ell m}(\iota, \varphi_0)} \quad (6)$$

This is identical in form to the time-domain expansion—the Fourier transform is linear, so it commutes with the spherical harmonic sum.

3.2 Intrinsic vs Extrinsic Parameters

A crucial advantage of the frequency domain is that extrinsic parameters enter in simple, analytic ways.

Intrinsic parameters $\boldsymbol{\lambda}$: These affect the *shape* of the waveform in $\mathcal{F}$ space:

$$\boldsymbol{\lambda} = \{\eta, \chi_{1z}, \chi_{2z}\} \quad (\text{aligned-spin}) \quad (7)$$

or more generally:

$$\boldsymbol{\lambda} = \{\eta, \chi_{1x}, \chi_{1y}, \chi_{1z}, \chi_{2x}, \chi_{2y}, \chi_{2z}\} \quad (\text{precessing}) \quad (8)$$

where $\eta = m_1 m_2 / (m_1 + m_2)^2$ is the symmetric mass ratio.

Note: The total mass M is *not* an intrinsic parameter for the emulator. The network learns functions of $\mathcal{F}$; the mass enters only when converting to physical frequency.

Extrinsic parameters: Each enters as a simple multiplicative or phase factor:

1. **Total mass** M : Converts $\mathcal{F} \rightarrow f$ and provides the amplitude prefactor. See §3.3.
2. **Luminosity distance** D_L : Scales amplitude as $\tilde{h} \propto 1/D_L$.
3. **Inclination ι and azimuth φ_0** : Enter only through ${}^{-2}Y_{\ell m}(\iota, \varphi_0)$ in (6).
4. **Coalescence time** t_c : Shifts phase linearly in frequency:

$$\tilde{h}_{\ell m}(f; t_c) = \tilde{h}_{\ell m}(f; t_c = 0) \times e^{-2\pi i f t_c} \quad (9)$$

5. **Coalescence phase** ϕ_c : Rotates mode phases:

$$\tilde{h}_{\ell m}(f; \phi_c) = \tilde{h}_{\ell m}(f; \phi_c = 0) \times e^{-im\phi_c} \quad (10)$$

6. **Polarisation angle** ψ : Mixes h_+ and $h_\times$ at the detector (not part of waveform generation).

Implication for emulation. The neural network learns only the *intrinsic mode shapes*:

$$\text{NN} : (\mathcal{F}, \boldsymbol{\lambda}) \mapsto \tilde{h}_{\ell m}(\mathcal{F}; \boldsymbol{\lambda}) \quad (11)$$

All extrinsic parameters are applied analytically at inference time. This reduces the effective parameter space from ~ 15 dimensions (full parameter estimation) to 3–7 dimensions (intrinsic only).

3.3 The Mass Scaling

The dependence on total mass deserves careful treatment. We *define* a dimensionless reduced waveform $H_{\ell m}$ by factoring out the mass and distance dependence:

$$\boxed{\tilde{h}_{\ell m}(f_{\text{phys}}) \equiv \frac{(GM/c^3)^2}{D_L} \times H_{\ell m}(\mathcal{F}; \boldsymbol{\lambda})} \quad (12)$$

where $\mathcal{F} = GMf_{\text{phys}}/c^3$ is the dimensionless frequency and $H_{\ell m}$ depends only on $\mathcal{F}$ and intrinsic parameters $\boldsymbol{\lambda}$.

The prefactor $(GM/c^3)^2/D_L$ has dimensions of time (consistent with $\tilde{h}$ being a Fourier transform). In geometric units ($G = c = 1$), this simplifies to M^2/D_L .

Consistency check. The Newtonian inspiral amplitude scales as $\tilde{A}_{22} \propto \mathcal{M}_c^{5/6} f^{-7/6}/D_L$ where $\mathcal{M}_c = M\eta^{3/5}$ is the chirp mass. Rewriting in terms of $\mathcal{F}$:

$$f^{-7/6} = M^{7/6}(Mf)^{-7/6}, \quad \mathcal{M}_c^{5/6} = M^{5/6}\eta^{1/2} \quad (13)$$

so $\tilde{A}_{22} \propto M^2(Mf)^{-7/6}\eta^{1/2}/D_L$, confirming that the M^2/D_L prefactor is absorbed leaving $H_{22} \propto (Mf)^{-7/6}\eta^{1/2}$.

For emulation: The network learns the dimensionless shape $H_{\ell m}(\mathcal{F}; \boldsymbol{\lambda})$. At inference:

1. Given physical parameters $(M, D_L, f_{\text{phys}})$, compute $\mathcal{F} = GMf_{\text{phys}}/c^3$
2. Evaluate $H_{\ell m}(\mathcal{F}; \boldsymbol{\lambda})$ from network
3. Apply prefactor: $\tilde{h}_{\ell m} = (GM/c^3)^2/D_L \times H_{\ell m}$

3.4 Amplitude–Phase Representation

Each mode is decomposed into amplitude and phase:

$$\boxed{\tilde{h}_{\ell m}(f; \boldsymbol{\lambda}) = \tilde{A}_{\ell m}(f; \boldsymbol{\lambda}) e^{i\Phi_{\ell m}(f; \boldsymbol{\lambda})}} \quad (14)$$

where $\tilde{A}_{\ell m}(f) \geq 0$ is the (real, positive) amplitude and $\Phi_{\ell m}(f)$ is the (real) phase.

Amplitude characteristics.

- Dynamic range: ~ 5 orders of magnitude from low frequency to ringdown
- Power-law behaviour in inspiral: $\tilde{A}_{\ell m} \propto f^{-7/6}$ for dominant mode
- Peak near merger frequency $\mathcal{F} \sim 0.02$
- Exponential decay in ringdown

Phase characteristics.

- Accumulates ~ 100 – 200 radians over typical observation band
- Dominated by chirp: $\Phi_{\ell m} \propto f^{-5/3}$ at low frequency (Newtonian)
- Must be continuous (unwrapped) for learning
- Mode phases related: $\Phi_{\ell m} \approx m \times \Phi_{22}/2$ to leading order

Practical note. The amplitude varies over many orders of magnitude while the phase varies over hundreds of radians. These disparate scales suggest different representations (see §5).

4 Symmetries and Constraints

4.1 Reality Condition

The polarisations $h_+(t)$ and $h_\times(t)$ are real-valued functions. Although the complex strain $h = h_+ - ih_\times$ is itself complex, the reality of its components implies:

$$\tilde{h}(-f) = \tilde{h}^*(f) \quad (15)$$

We need only model positive frequencies $f > 0$; negative frequency content is determined by this conjugate relation. The standard matched-filter inner product uses only positive frequencies, with the “4 Re” prefactor accounting for the negative-frequency contribution.

4.2 Conjugate Mode Symmetry (Non-Precessing)

For non-precessing systems with *aligned* spins, the waveform has a reflection symmetry about the orbital plane. In the *time domain*, this implies:

$$h_{\ell, -m}(t) = (-1)^\ell h_{\ell m}^*(t) \quad (16)$$

Taking the Fourier transform:

$$\tilde{h}_{\ell, -m}(f) = (-1)^\ell \tilde{h}_{\ell m}^*(-f) \quad (17)$$

This relates the $m < 0$ mode at frequency f to the $m > 0$ mode at frequency $-f$.

Implication for emulation. For quasi-circular non-precessing binaries, the $m > 0$ and $m < 0$ modes are not independent—they are related by the symmetry (17). We therefore emulate only $\tilde{h}_{\ell m}(f)$ for $m > 0$ and $f > 0$.

Strain reconstruction. Within the standard matched-filter inner product (one-sided, with the “4 Re” prefactor), the strain at positive frequency is computed as:

$$\tilde{h}(f > 0) = \sum_{\ell} \sum_{m>0} \tilde{h}_{\ell m}(f) {}^{-2}Y_{\ell m}(\iota, \varphi_0) \quad (18)$$

The negative-frequency contribution (and hence the $m < 0$ mode content) is accounted for by the inner product convention. When a two-sided Fourier representation is required (e.g., for IFFT to time domain), one first assembles $\tilde{h}(f > 0)$ using (18), then constructs $\tilde{h}(f < 0) = \tilde{h}^*(-f)$ from the full-strain reality condition.

For IMRPhenomXHM, the relevant modes are: (2, 2), (2, 1), (3, 3), (3, 2), (4, 4).

4.3 Mode Hierarchy

Not all modes contribute equally:

Mode	Relative Amplitude	Peak Frequency
(2, 2)	1 (reference)	$\mathcal{F} \approx 0.02$
(2, 1)	$\sim 0.1\text{--}0.3$	$\mathcal{F} \approx 0.01$
(3, 3)	$\sim 0.05\text{--}0.3$	$\mathcal{F} \approx 0.03$
(3, 2)	$\sim 0.01\text{--}0.1$	$\mathcal{F} \approx 0.02$
(4, 4)	$\sim 0.01\text{--}0.1$	$\mathcal{F} \approx 0.04$

Table 1: Approximate mode amplitudes relative to (2, 2), depending on mass ratio and spins.

Higher modes become more important for:

- Unequal masses ($q \neq 1$)
- Edge-on inclinations ($\iota \approx \pi/2$)
- Higher total masses (where fewer inspiral cycles are in band)

4.4 Phase Conventions and Degeneracies

Several phase parameters are partially degenerate:

- **Reference phase** ϕ_{ref} : The phase of the (2, 2) mode at reference frequency
- **Coalescence phase** ϕ_c : The orbital phase at coalescence
- **Azimuthal angle** φ_0 : Viewing angle in orbital plane

For data analysis, only certain combinations are measurable. For emulation, we fix a convention:

$$\Phi_{22}(\mathcal{F}_{\text{ref}}) = 0 \quad (19)$$

This removes the phase freedom in training data.

5 Representation Choices for Neural Networks

5.1 Amplitude Representation

The amplitude $\tilde{A}_{\ell m}(\mathcal{F})$ spans ~ 5 orders of magnitude. Direct representation is challenging for neural networks. Options:

Option A: Log-amplitude.

$$\text{Output : } \log \tilde{A}_{\ell m}(\mathcal{F}) \quad (20)$$

Pros: Handles dynamic range naturally, errors become relative

Cons: Undefined when $\tilde{A}_{\ell m} \rightarrow 0$ (mode zeros), log-errors weight low amplitudes heavily

Option B: Normalised amplitude.

$$\text{Output : } \frac{\tilde{A}_{\ell m}(\mathcal{F})}{\tilde{A}_{22}^{\text{ref}}(\mathcal{F})} \quad (21)$$

where $\tilde{A}_{22}^{\text{ref}}$ is a reference (2, 2) amplitude (e.g., at $\eta = 0.25$, $\chi = 0$).

Pros: Order-unity outputs, physically motivated

Cons: Requires reference waveform, extra preprocessing

Option C: Post-Newtonian subtraction.

$$\text{Output : } \frac{\tilde{A}_{\ell m}(\mathcal{F})}{\tilde{A}_{\ell m}^{\text{PN}}(\mathcal{F})} \quad (22)$$

where $\tilde{A}_{\ell m}^{\text{PN}}$ is the leading-order post-Newtonian amplitude.

Pros: Network learns only the “correction” to known physics

Cons: PN diverges at merger, need transition handling

Recommendation : Log-amplitude (Option A) for its simplicity, with careful handling of mode zeros if they occur in the training data.

5.2 Phase Representation

The phase $\Phi_{\ell m}(\mathcal{F})$ accumulates hundreds of radians. Options:

Option A: Direct (unwrapped) phase.

$$\text{Output : } \Phi_{\ell m}(\mathcal{F}) \quad (23)$$

Pros: Simple, direct

Cons: Large dynamic range, errors in radians may be acceptable but scale poorly

Option B: Phase derivative (instantaneous frequency).

$$\text{Output : } \frac{d\Phi_{\ell m}}{d(\mathcal{F})} = \omega_{\ell m}(\mathcal{F}) \quad (24)$$

Reconstruct phase by integration: $\Phi_{\ell m}(\mathcal{F}) = \int^{\mathcal{F}} \omega_{\ell m} d(\mathcal{F}')$

Pros: Physical interpretation (related to time-to-merger)

Cons: Large dynamic range at low frequency ($d\Phi/d(Mf) \propto (Mf)^{-8/3}$ in inspiral), integration accumulates errors, boundary condition needed

Option C: Residual after PN subtraction.

$$\text{Output : } \Phi_{\ell m}(\mathcal{F}) - \Phi_{\ell m}^{\text{PN}}(\mathcal{F}; \boldsymbol{\lambda}) \quad (25)$$

where $\Phi_{\ell m}^{\text{PN}}$ is the post-Newtonian phase to high order.

Pros: Network learns small corrections, PN is well-understood

Cons: PN coefficients depend on parameters, need to evaluate PN at inference

Option D: TaylorF2 residual.

$$\text{Output : } \Phi_{\ell m}(\mathcal{F}) - \Phi_{\ell m}^{\text{TF2}}(\mathcal{F}; \boldsymbol{\lambda}) \quad (26)$$

where $\Phi_{\ell m}^{\text{TF2}}$ is the TaylorF2 phase (analytic PN in frequency domain).

Pros: TaylorF2 is fast to evaluate, captures bulk of inspiral phase

Cons: Breaks down at merger, need smooth transition

Recommendation : For initial implementation, direct phase (Option A) with unwrapping. For production accuracy, consider PN residual (Option C/D) to reduce the dynamic range the network must capture.

5.3 Frequency Grid

The network can operate on:

Fixed grid in $\mathcal{F}$. Train and evaluate on a fixed set of geometric frequencies $\{\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_N\}$.

Grid spacing: The waveform has features on different scales:

- Inspiral: Slowly varying, coarse grid sufficient
- Merger: Rapid variation, fine grid needed
- Ringdown: Quasi-periodic, moderate grid

A log-spaced grid is natural, matching the power-law inspiral behaviour:

$$\mathcal{F}_k = \mathcal{F}_{\min} \times \left(\frac{\mathcal{F}_{\max}}{\mathcal{F}_{\min}} \right)^{k/(N-1)}, \quad k = 0, \dots, N-1 \quad (27)$$

giving N points from $\mathcal{F}_{\min}$ to $\mathcal{F}_{\max}$.

Continuous evaluation. Instead of a fixed grid, output basis coefficients that allow evaluation at any $\mathcal{F}$:

$$\tilde{A}_{\ell m}(\mathcal{F}) = \sum_{k=1}^K c_k^{(A)}(\boldsymbol{\lambda}) B_k(\mathcal{F}) \quad (28)$$

where $B_k(\mathcal{F})$ are basis functions (splines, PCA modes, etc.).

Pros: Arbitrary resolution at inference, interpolation is exact within basis

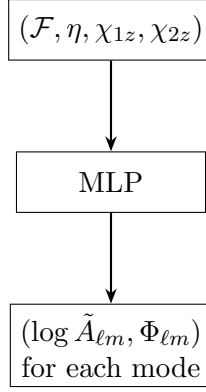
Cons: More complex architecture, basis must be chosen carefully

Recommendation : For initial implementation, use a fixed log-spaced grid with ~ 1000 points. Interpolate for intermediate frequencies if needed.

6 Neural Network Architecture

6.1 Direct Mapping (Per-Frequency)

The simplest architecture evaluates each frequency independently:



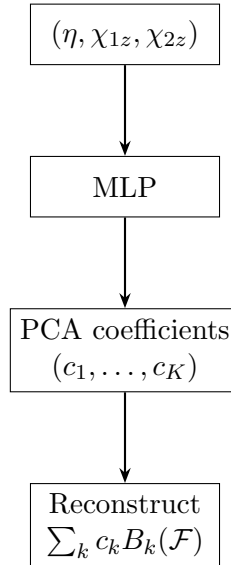
Pros: Simple, embarrassingly parallel over frequencies

Cons: Doesn't exploit correlations across frequency, large network needed

6.2 PCA + MLP (CosmoPower Architecture)

Following CosmoPower [2]:

1. **Training data:** Generate waveforms on fixed grid for many parameter values
2. **PCA:** Compute principal components across training set
3. **Network:** MLP maps parameters to PCA coefficients
4. **Reconstruction:** Combine PCA modes to get waveform

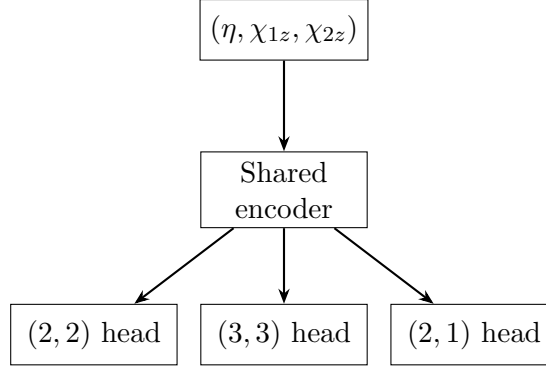


Pros: Exploits waveform correlations, compact representation, CosmoPower-validated

Cons: Separate PCA for each quantity (amplitude, phase, per mode), fixed frequency grid

6.3 Mode-Coupled Architecture

Higher modes are correlated with the (2, 2) mode (they come from the same binary). A coupled architecture can exploit this:



Pros: Shared features, potentially smaller total network

Cons: More complex, mode dependencies may not help much

6.4 Recommended Starting Point

For initial development, we recommend:

1. **PCA + MLP architecture** (CosmoPower-style)
2. **Separate models per mode** initially, unify later if beneficial
3. **Log-amplitude** and **direct phase** outputs
4. **Log-spaced grid** in $\mathcal{F}$
5. **Speculator activation function** $\sigma(x) = x \cdot \text{sigmoid}(x)$ [3]

7 The Precessing Case

For precessing systems, frequency-domain modelling is more complex. The time-domain coprecessing frame decomposition does not have a direct frequency-domain analog because the Euler angles $\alpha(t), \beta(t), \gamma(t)$ are functions of time, not frequency.

7.1 The Twisting-Up Approximation

IMRPhenomXPHM and similar models use the “twisting-up” approach:

1. Compute non-precessing (aligned-spin) modes in frequency domain
2. Apply frequency-dependent rotation to account for precession

The key approximation is the **stationary phase assumption**: each frequency component f corresponds to a single time $t(f)$ via the time-frequency relation. In practice, the mapping $t(f)$ is derived from the *dominant* (2,2) mode and then applied to all modes—an approximation that becomes less accurate for subdominant modes with different frequency content. The Euler angles at that time, $\alpha(t(f)), \beta(t(f)), \gamma(t(f))$, are then used to rotate the modes:

$$\tilde{h}_{\ell m}(f) = \sum_{m'=-\ell}^{+\ell} D_{mm'}^{\ell}(\alpha(t(f)), \beta(t(f)), \gamma(t(f))) \tilde{h}_{\ell m'}^{\text{AS}}(f) \quad (29)$$

where $\tilde{h}^{\text{AS}}$ are aligned-spin modes.

7.2 Limitations of Frequency-Domain Precession

The twisting-up approach has known limitations:

- **Mode mixing:** The Wigner rotation redistributes power among modes with different m' —this is the *intended* effect. The limitation is that the rotation is evaluated at a single $t(f)$ per frequency, whereas true precession creates sidebands at multiple frequencies.
- **Multiple timescales:** Strong precession violates stationary phase—a single frequency f may correspond to multiple times t , or the precession timescale may be comparable to the orbital timescale.
- **Transitional precession:** Near $L \sim S$, the frame can flip rapidly, causing the Euler angles to vary on timescales faster than the $(2, 2)$ mode frequency evolution.

For highly precessing systems, time-domain modelling (as in the companion document) may be more appropriate.

7.3 Emulation Strategy for Precessing Waveforms

Two approaches for emulating precessing frequency-domain waveforms:

Option 1: Emulate the twisted output. Treat the full $\tilde{h}_{\ell m}(f; \boldsymbol{\lambda})$ including precession as the target. The network learns the combined effect of mode evolution and precession.

Pros: Simple, single network

Cons: Must learn oscillatory structure from precession, larger network needed

Option 2: Emulate components separately. Train separate networks for:

- Aligned-spin modes $\tilde{h}_{\ell m'}^{\text{AS}}(f)$
- Euler angles as functions of frequency $\alpha(f), \beta(f), \gamma(f)$

Combine via Wigner rotation at inference.

Pros: Modular, aligned-spin emulator can be reused

Cons: Two networks, angle-frequency relation approximate

Recommendation : For this project, focus on non-precessing (aligned-spin) waveforms first. Precession can be added later following Option 2, building on the aligned-spin emulator.

8 Training and Validation

8.1 Training Data Generation

Parameter sampling.

- **Symmetric mass ratio** $\eta \in [0.05, 0.25]$ (mass ratios $q \in [1, 18]$)
- **Spin components** $\chi_{1z}, \chi_{2z} \in [-0.99, 0.99]$
- **Sampling:** Latin hypercube or Sobol sequence for space-filling
- **Training set size:** $\sim 10^4$ – 10^5 waveforms

Waveform generation. Use LALSimulation to generate IMRPhenomXHM waveforms:

```
from lal import MSUN_SI
import lalsimulation as lalsim

hp, hc = lalsim.SimIMRPhenomXHM(
    m1_SI, m2_SI, S1z, S2z, # masses and spins
    distance, inclination, # extrinsic (set to reference)
    phiRef, deltaF, f_min, # frequency grid
    f_ref, lalParams        # reference and options
)
```

Time and phase alignment. Critical: All training waveforms must be consistently aligned before extracting amplitude and phase. A time shift t_0 in the time domain corresponds to a phase term $e^{-2\pi i f t_0}$ in the frequency domain, adding a linear component $2\pi f t_0$ to the phase. If waveforms have different effective t_0 values (due to different merger times relative to the data window), the phase $\Phi(\mathcal{F})$ will contain a parameter-dependent linear trend that is extremely difficult for the network to learn.

Alignment procedure:

1. Generate waveforms with a common reference (e.g., coalescence at $t_c = 0$)
2. Alternatively, post-process by fitting and removing a linear phase: $\Phi(f) \rightarrow \Phi(f) - 2\pi f t_0 - \phi_0$
3. Fix the (2,2) phase at the reference frequency: $\Phi_{22}(\mathcal{F}_{\text{ref}}) = 0$

Without proper alignment, the phase emulator will likely fail to converge.

Data format. Store in HDF5:

- Frequency grid: $\mathcal{F}$ values (common to all waveforms)
- Per waveform: parameters $(\eta, \chi_{1z}, \chi_{2z})$, amplitudes, phases for each mode
- Metadata: LAL version, waveform approximant, alignment convention used

8.2 Loss Functions

Mean squared error. On log-amplitude and phase:

$$\mathcal{L}_{\text{MSE}} = \sum_{\ell m} \left[\left\| \log \tilde{A}_{\ell m}^{\text{pred}} - \log \tilde{A}_{\ell m}^{\text{true}} \right\|^2 + w_{\Phi} \left\| \Phi_{\ell m}^{\text{pred}} - \Phi_{\ell m}^{\text{true}} \right\|^2 \right] \quad (30)$$

where w_{Φ} balances amplitude and phase contributions.

Complex MSE. Directly on the complex waveform:

$$\mathcal{L}_{\text{complex}} = \sum_{\ell m} \left\| \tilde{h}_{\ell m}^{\text{pred}} - \tilde{h}_{\ell m}^{\text{true}} \right\|^2 \quad (31)$$

This automatically balances amplitude and phase.

8.3 Validation: The Mismatch

The standard accuracy metric for gravitational waveforms is the **mismatch**:

$$\mathcal{M} = 1 - \max_{t_c, \phi_c} \frac{\langle h_1 | h_2 \rangle}{\sqrt{\langle h_1 | h_1 \rangle \langle h_2 | h_2 \rangle}} \quad (32)$$

where the noise-weighted inner product is:

$$\langle h_1 | h_2 \rangle = 4 \operatorname{Re} \int_{f_{\text{low}}}^{f_{\text{high}}} \frac{\tilde{h}_1(f) \tilde{h}_2^*(f)}{S_n(f)} df \quad (33)$$

and $S_n(f)$ is the detector noise power spectral density.

Target accuracy. For parameter estimation, mismatches $\lesssim 10^{-3}$ are typically required. This corresponds to:

- Amplitude errors: $\lesssim 1\%$ relative
- Phase errors: $\lesssim 0.1$ radians

Mismatch as loss function. The mismatch can be used directly as a loss function, but requires:

- Differentiable inner product implementation
- Handling of maximisation over t_c, ϕ_c
- Choice of reference PSD $S_n(f)$

The maximisation over t_c is typically performed via inverse FFT of the frequency-domain inner product integrand, which finds the optimal time shift efficiently. This operation is differentiable (FFTs are linear), enabling end-to-end training with mismatch loss if desired.

For initial training, MSE is simpler. Mismatch is better suited for validation and fine-tuning.

9 Comparison with Time-Domain Framework

Aspect	Time Domain	Frequency Domain
Natural domain variable	t/M	Mf
Coprocessing frame needed	Yes (precessing)	No (approximated)
Euler angle dynamics	Learned	Derived from $t(f)$
Extrinsic parameters	Mixed in	Analytic factors
Phase accumulation	Implicit	Explicit (many radians)
Mode mixing	Via Wigner rotation	Via twisting-up
Direct data analysis use	Requires FFT	Yes
Best for	Precessing, eccentric	Aligned-spin, circular

Table 2: Comparison of time-domain and frequency-domain emulation frameworks.

The two frameworks are complementary:

- **Frequency domain:** Simpler for non-precessing waveforms, direct use in data analysis, cleaner mass scaling
- **Time domain:** More natural for precession, handles multiple timescales, needed for eccentric systems

For this project (IMRPhenomXHM emulation), the frequency domain is the natural choice.

10 Summary

The frequency-domain framework for waveform emulation is:

$$\tilde{h}(f; \boldsymbol{\lambda}; \iota, \varphi_0) = \sum_{\ell, m} \tilde{h}_{\ell m}(f; \boldsymbol{\lambda})^{-2} Y_{\ell m}(\iota, \varphi_0) \quad (34)$$

$$\tilde{h}_{\ell m}(f; \boldsymbol{\lambda}) = \tilde{A}_{\ell m}(f; \boldsymbol{\lambda}) e^{i\Phi_{\ell m}(f; \boldsymbol{\lambda})} \quad (35)$$

$$\text{NN} : (\mathcal{F}, \boldsymbol{\lambda}) \mapsto (\log \tilde{A}_{\ell m}, \Phi_{\ell m}) \quad \text{for each mode} \quad (36)$$

Key points:

1. The dimensionless frequency $\mathcal{F} = Mf$ is the natural domain variable—total mass is not an intrinsic parameter.
2. Extrinsic parameters (distance, coalescence time/phase, inclination) enter analytically via simple multiplicative factors.
3. For non-precessing systems, conjugate symmetry halves the number of modes to emulate.
4. Log-amplitude and direct phase are suitable output representations for neural networks.
5. The PCA + MLP architecture (CosmoPower-style) is a proven approach.
6. Mismatch is the standard validation metric; target $\mathcal{M} < 10^{-3}$.
7. Precessing waveforms can be added via separate Euler angle emulation (twisting-up), building on the aligned-spin base.

Recommended starting point: Emulate IMRPhenomXHM modes (2, 2), (2, 1), (3, 3), (3, 2), (4, 4) for aligned-spin parameter space $(\eta, \chi_{1z}, \chi_{2z})$ using PCA + MLP architecture with log-amplitude and unwrapped phase outputs.

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